

Ontology Engineering

Session 6

Project Overview

Vienna Convention on
Road Signs and Signals
(Part I, Annex 1)

LLM-based ontology development

road-sign ontology
in OWL 2
(RDF/XML Syntax)

taxonomy or road-sign classes

object and data properties

property characteristics

definitions
(necessary and sufficient conditions)

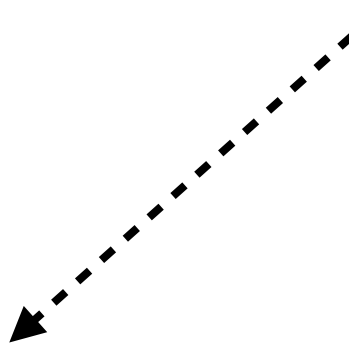
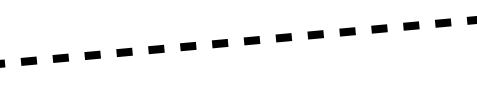
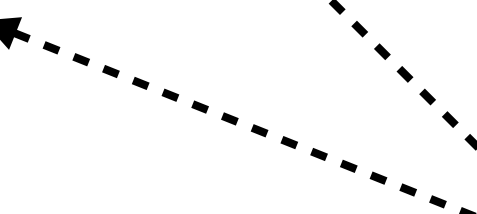
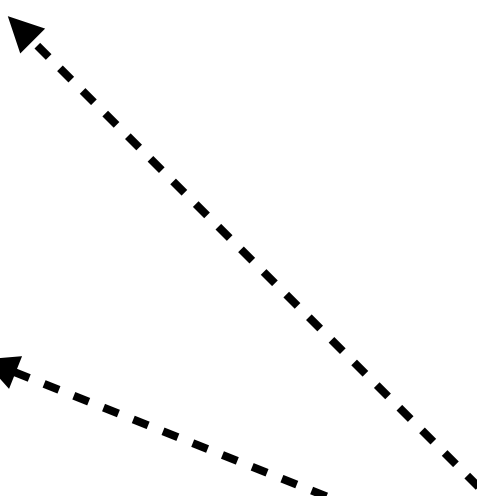
Class
Alignment

Mapillary Traffic Sign Dataset
Knowledge Graph
(MTSD.ttl)

Competency
Questions

LLM-based
coding

Python script
DL Queries
SPARQL



The Road-Sign Ontology

OntologyID (Anonymous-2) : [/Users/schorlemmer/Documents/UAB/KR2026/project/final project/roadsignonto.rdf.xml]

OntologyID(Anonymous-2)

RoadSign > PrioritySign > B2_Stop

Active ontology x Entities x Individuals by class x DL Query x SPARQL Query x

Annotation properties Datatypes Individuals

Classes Object properties Data properties

Class hierarchy: B2_Stop

Asserted

- owl:Thing
 - Colour
 - FacilityType
 - HazardType
 - LaneType
 - Model
 - Orientation
 - RegulationType
 - RoadSign
 - AdditionalPanel
 - DangerWarningSign
 - DirectionPositionIndicationSign
 - InformationFacilityServiceSign
 - MandatorySign
 - PrioritySign
 - B1_GiveWay
 - B2_Stop**
 - B3_PriorityRoad
 - B4_EndOfPriority
 - B5_PriorityForOncomingTraffic
 - B6_PriorityOverOncomingTraffic
 - ProhibitoryOrRestrictiveSign
 - SpecialRegulationSign
 - Shape
 - SizeCategory
 - Symbol
 - UserCategory

B2_Stop — http://example.org/roadsigns#B2_Stop

Annotations Usage

Annotations: B2_Stop

Annotations +

Description: B2_Stop

Equivalent To +

- PrioritySign
 - and (hasBorderColour value White)
 - and (hasGroundColour value Red)
 - and (hasShape value Octagon)
 - and (hasText value "STOP")

SubClass Of +

- PrioritySign

General class axioms +

SubClass Of (Anonymous Ancestor)

- RoadSign
 - and (indicatesRegulationType value PriorityRule)
- AdditionalPanel or DangerWarningSign or DirectionPositionIndicationSign or InformationFacilityServiceSign or MandatorySign or PrioritySign or ProhibitoryOrRestrictiveSign or SpecialRegulationSign

Instances +

Target for Key +

Disjoint With +

- B3_PriorityRoad, B4_EndOfPriority, B5_PriorityForOncomingTraffic, B6_PriorityOverOncomingTraffic, B1_GiveWay

Disjoint Union Of +

To use the reasoner click Reasoner > Start reasoner ☒ Show Inferences

The MTDS Knowledge Graph

```
MTSD.ttl ~

<https://example.org/mtsd/image/--7fWq6WjZM8L1eUSuv0EA>
  a mtsd:Image ;
  mtsd:sourceFile "--7fWq6WjZM8L1eUSuv0EA.json" ;
  mtsd:width "4160"^^xsd:integer ;
  mtsd:height "3120"^^xsd:integer ;
  mtsd:isPano "false"^^xsd:boolean ;
  mtsd:hasObject <https://example.org/mtsd/object/--7fWq6WjZM8L1eUSuv0EA/iw8oed2vtpa46rfvmki2wk>,
    <https://example.org/mtsd/object/--7fWq6WjZM8L1eUSuv0EA/pyiahdn8h67q6s4lt90bn3>,
    <https://example.org/mtsd/object/--7fWq6WjZM8L1eUSuv0EA/7p50jh0w90ybtkau7w154>,
    <https://example.org/mtsd/object/--7fWq6WjZM8L1eUSuv0EA/8rynh9ttz4tubjvhnbl1a2o> .

<https://example.org/mtsd/object/--7fWq6WjZM8L1eUSuv0EA/iw8oed2vtpa46rfvmki2wk>
  a mtsd:TrafficSignObject ;
  mtsd:key "iw8oed2vtpa46rfvmki2wk" ;
  rdfs:label "regulatory--keep-right--g1" ;
  mtsd:xmin "2187.65625"^^xsd:decimal ;
  mtsd:ymin "1669.6875"^^xsd:decimal ;
  mtsd:xmax "2225.234375"^^xsd:decimal ;
  mtsd:ymax "1707.7734375"^^xsd:decimal ;
  mtsp:barrier "false"^^xsd:boolean ;
  mtsp:occluded "false"^^xsd:boolean ;
  mtsp:out-of-frame "false"^^xsd:boolean ;
  mtsp:exterior "false"^^xsd:boolean ;
  mtsp:ambiguous "false"^^xsd:boolean ;
  mtsp:included "false"^^xsd:boolean ;
  mtsp:direction-or-information "false"^^xsd:boolean ;
  mtsp:highway "false"^^xsd:boolean ;
  mtsp:dummy "false"^^xsd:boolean .

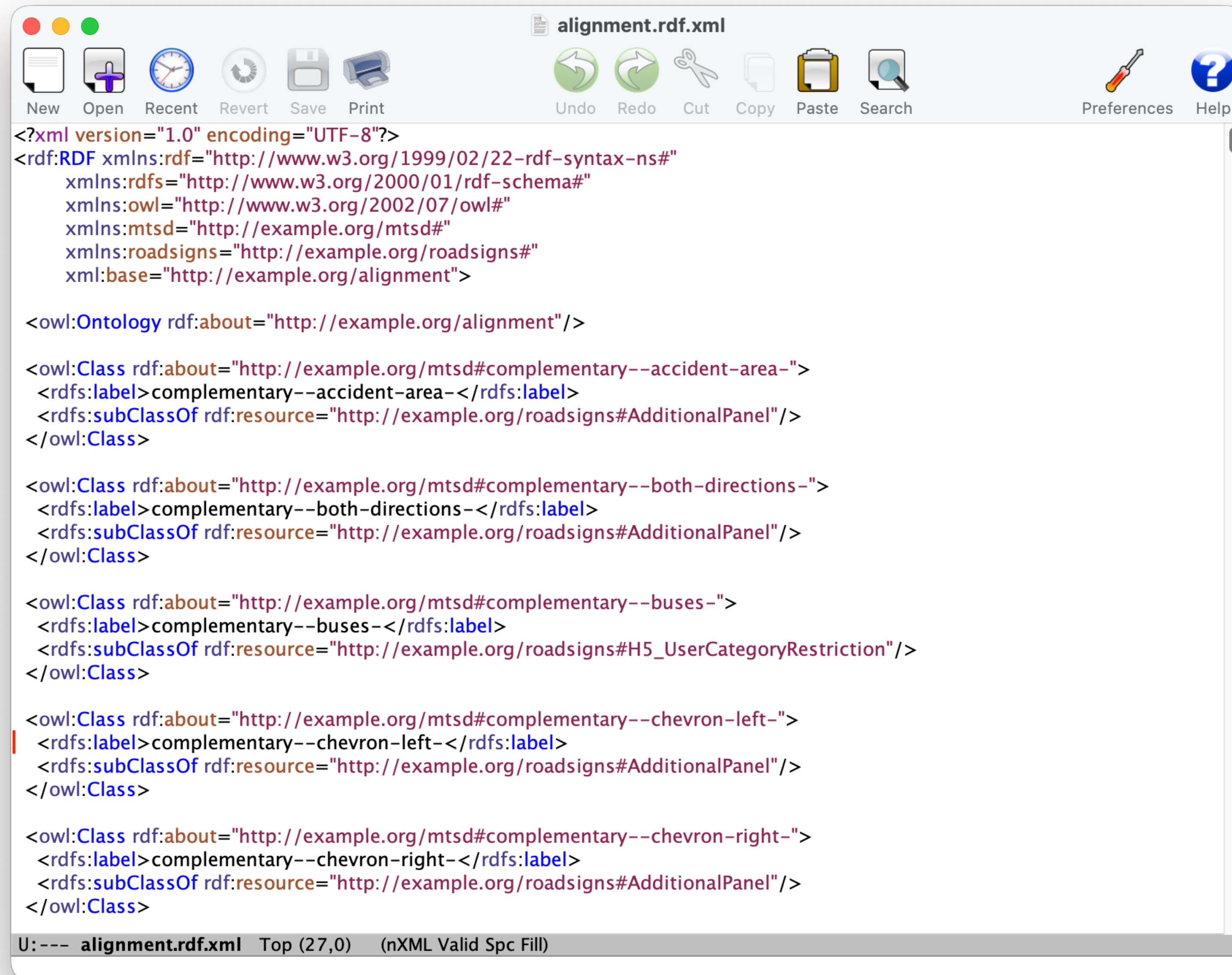
<https://example.org/mtsd/object/--7fWq6WjZM8L1eUSuv0EA/pyiahdn8h67q6s4lt90bn3>
  a mtsd:TrafficSignObject ;
  mtsd:key "pyiahdn8h67q6s4lt90bn3" ;
  rdfs:label "regulatory--priority-over-oncoming-vehicles--g1" ;
  mtsd:xmin "2511.640625"^^xsd:decimal ;
  mtsd:ymin "1391.66015625"^^xsd:decimal ;
  mtsd:xmax "2623.359375"^^xsd:decimal ;
  mtsd:ymax "1508.96484375"^^xsd:decimal ;
  mtsp:barrier "false"^^xsd:boolean ;
  mtsp:occluded "false"^^xsd:boolean ;
  mtsp:out-of-frame "false"^^xsd:boolean ;
  mtsp:exterior "false"^^xsd:boolean ;
  mtsp:ambiguous "false"^^xsd:boolean ;
  mtsp:included "false"^^xsd:boolean ;
  mtsp:direction-or-information "false"^^xsd:boolean ;
  mtsp:highway "false"^^xsd:boolean ;
  mtsp:dummy "false"^^xsd:boolean .
```


Class Alignment with MTDS

The alignment should be stated with class equivalences using **owl:equivalentClass** when they refer to the same class of road signs, or with class subsumption using **rdfs:subClassOf** when an exact equivalence cannot be inferred.

- **exact match** (owl:equivalentClass):
information--bus-stop- $\equiv$ E15_BusStop ()
- **partial match** (rdfs:subClassOf):
regulatory--maximum-speed-limit-120- $\sqsubseteq$ C14_MaximumSpeed

Class Alignment with MTDS



The screenshot shows a text editor window titled "alignment.rdf.xml" with a standard macOS-style toolbar at the top. The toolbar includes icons for New, Open, Recent, Revert, Save, Print, Undo, Redo, Cut, Copy, Paste, Search, Preferences, and Help. The main text area contains the following XML code:

```
<?xml version="1.0" encoding="UTF-8"?>
<rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#"
  xmlns:rdfs="http://www.w3.org/2000/01/rdf-schema#"
  xmlns:owl="http://www.w3.org/2002/07/owl#"
  xmlns:mtsd="http://example.org/mtsd#"
  xmlns:roadsigns="http://example.org/roadsigns#"
  xml:base="http://example.org/alignment">

  <owl:Ontology rdf:about="http://example.org/alignment"/>

  <owl:Class rdf:about="http://example.org/mtsd#complementary--accident-area-">
    <rdfs:label>complementary--accident-area-</rdfs:label>
    <rdfs:subClassOf rdf:resource="http://example.org/roadsigns#AdditionalPanel"/>
  </owl:Class>

  <owl:Class rdf:about="http://example.org/mtsd#complementary--both-directions-">
    <rdfs:label>complementary--both-directions-</rdfs:label>
    <rdfs:subClassOf rdf:resource="http://example.org/roadsigns#AdditionalPanel"/>
  </owl:Class>

  <owl:Class rdf:about="http://example.org/mtsd#complementary--buses-">
    <rdfs:label>complementary--buses-</rdfs:label>
    <rdfs:subClassOf rdf:resource="http://example.org/roadsigns#H5_UserCategoryRestriction"/>
  </owl:Class>

  <owl:Class rdf:about="http://example.org/mtsd#complementary--chevron-left-">
    <rdfs:label>complementary--chevron-left-</rdfs:label>
    <rdfs:subClassOf rdf:resource="http://example.org/roadsigns#AdditionalPanel"/>
  </owl:Class>

  <owl:Class rdf:about="http://example.org/mtsd#complementary--chevron-right-">
    <rdfs:label>complementary--chevron-right-</rdfs:label>
    <rdfs:subClassOf rdf:resource="http://example.org/roadsigns#AdditionalPanel"/>
  </owl:Class>

</rdf:RDF>
```

The status bar at the bottom of the window displays "U: --- alignment.rdf.xml Top (27,0) (nXML Valid Spc Fill)".

Competency Questions

- **CQs about the taxonomy of road signs**

- What are the primary kinds of road signs distinguished in the Vienna Convention?
- Is a priority sign considered a type of road sign?

- **CQs about road sign properties**

- Which border colours can circle-shaped road signs have?
- What are the required background and border colours for a model B2a "STOP" sign?
- What is the standard measurement for the side of a normal-sized danger warning sign?

- **CQs about the pragmatics of road signs**

- Which danger warning symbol (A11a or A11b) must be used to warn of falling rocks?
- What specific symbol is used on sign A15 to warn of "Cattle or other animals crossing"?
- Which road signs are relevant to pedestrians?

- **CQs relevant to the particular dataset**

- How many road signs are depicted in the image IMG_20230314_49094.jpg?
- Which images contain a “COMPULSORY MINIMUM SPEED” sign?
- What percentage of road signs depicted in the image repository are prohibitory signs?
- Is there any road sign in the image repository that informs about a restriction applicable to lorries or other large vehicles?
- What is the maximum speed shown on most speed limit signs?

Coding the CQs

- **Competency Questions:**
 - by way of **DL queries** (to be pasted into Protégé's 'DL Query' tab)
 - by way of **SPARQL queries** (only when the semantic reasoner is not required)
 - by way of **Python scripts** (using Owlready2)

Example (DL Query)

- Which road signs are relevant to pedestrians?

The screenshot shows the OntologyID web application interface. The browser address bar displays the URL: `OntologyID(Anonymous-2) : [/Users/schorlemmer/Documents/UAB/KR2026/project/final project/roadsignonto.rdf.xml]`. The application has several tabs: "Active ontology", "Entities", "Individuals by class", "DL Query", and "SPARQL Query". The "DL Query" tab is active.

Class hierarchy: A tree view on the left shows the ontology structure. The root is `owl:Thing`, which includes classes like `Colour`, `FacilityType`, `HazardType`, `LaneType`, `Model`, `Orientation`, `RegulationType`, `RoadSign`, `Shape`, `SizeCategory`, `Symbol`, and `UserCategory`. The `RoadSign` class is expanded, showing subclasses like `AdditionalPanel`, `DangerWarningSign`, `DirectionPositionIndicationSign`, `InformationFacilityServiceSign`, `MandatorySign`, `PrioritySign`, `ProhibitoryOrRestrictiveSign`, and `SpecialRegulationSign`.

DL query: The query input field contains the expression: `RoadSign and (appliesToUserCategory value Pedestrian)`. Below the input field are buttons for "Execute" and "Add to ontology".

Query results: The results section shows "Subclasses (5 of 5)" of the query. The results are:

- `A12_PedestrianCrossingAhead`
- `A13_Children`
- `D5_Footpath`
- `E12_PedestrianCrossing`
- `owl:Nothing`

Query for: A list of checkboxes on the right allows filtering the results:

- ☐ Direct superclasses
- ☐ Superclasses
- ☐ Equivalent classes
- ☐ Direct subclasses
- ☒ Subclasses
- ☐ Instances

Result filters: At the bottom right, there are checkboxes for "Reasoner active" (checked) and "Show Inferences" (checked).

Example (SPARQL)

- What are the primary kinds of road signs distinguished in the Vienna Convention?

```
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
```

```
PREFIX rs: <http://example.org/roadsigns#>
```

```
SELECT ?primaryKind
```

```
WHERE {
```

```
    ?primaryKind rdfs:subClassOf rs:RoadSign .
```

```
}
```

Example (Python)

- What percentage of road signs depicted in the image repository are prohibitory signs?

```
prohibitory_count = sum(
    count
    for label, count in mtsd.label_counts.items()
    if alignment.label_is_aligned_to(label, RS.ProhibitoryOrRestrictiveSign)
)
aligned_count = sum(
    count
    for label, count in mtsd.label_counts.items()
    if alignment.label_to_road_classes.get(normalise_mtsd_label(label))
)
print("11. Percentage of depicted road signs that are prohibitory/restrictive:")
print(f"    Against all MTSD objects: {prohibitory_count / mtsd.total_objects * 100:.2f}%")
print(f"    Against aligned MTSD objects: {prohibitory_count / aligned_count * 100:.2f}%")
```


Project Sumbission and Defence

- Submission deadline: **31 May 2026, 23:59 h**
- Defence: **8 June 2026, 15:00-17:00 h, Aula d'Informàtica A**
- Methodology:
 - Team defence – each team works on a single computer, all members are present
 - Editing your ontology with Protégé according to a given requirement (e.g. adding new classes with their definitions)
 - Coding competency questions and checking them against your ontology and a fragment of the MTDS knowledge graph (as a DL Query posted into Protégé).
 - Additional questions to selected groups
 - No access to the Internet